

Chapter 9

Potential Flow over Finite Wings

9.1. Prandtl's Classical Lifting-line Theory

In the analysis techniques, which we have studied to this point, we have been concerned with the aerodynamic properties of airfoils, which are the same as the aerodynamic properties of wings with infinite span. A wing of constant cross-section and infinite span would have no variation in aerodynamic forces in the spanwise direction. The aerodynamic forces per unit span acting on such a wing, at any point along the span, would be the same as those for an airfoil of the same cross-section. Airfoils, and thus wings of infinite span, are synthesized using vortex sheets that are made up of straight vortex filaments that extend to $\pm\infty$ in the direction of span. The vortex strength can vary over the sheet as we move in a chordwise direction from one vortex filament to another. However, there is no variation in vortex strength as we move along a vortex filament in the direction of span.

Any real wing, of course, must have finite span. At the tips of a finite wing, the air on the lower surface of the wing comes in direct contact with the air on the upper surface of the wing. Thus, at the tips of a finite wing, the pressure difference between the upper and lower surface must always go to zero. As a result, the lift on any finite wing must go to zero at the wingtips. The lift distribution on a finite wing is shown schematically in Fig. 9.1.1.

The pressure difference between the upper and lower surface of a finite wing is reduced near the wingtips, because some of the air from the high-pressure region below the wing spills outward, around the wingtip, and back inward toward the low-pressure region above the wing. Thus, while the flow around an infinite wing is entirely in the plane of the airfoil section, the flow around a finite wing is three-dimensional, with an additional component in the spanwise direction. The air below the wing flows outward toward the wingtip and the air above the wing flows inward toward the root, as shown in Fig 9.1.2. Where the flow from the upper and lower surfaces combine at the trailing edge, the difference in spanwise velocity generates a trailing vortex sheet.

Since the lift on a finite wing always goes to zero at the wingtips, if the wing develops finite lift, the lift per unit span must vary in the direction of span. This is an important concept when combined with the circulation theory of lift and Helmholtz's vortex theorems. The circulation theory of lift requires that the lift per unit span acting at any cross-section of a wing is proportional to the circulation around the wing at that cross-section (see App. F). Thus, since the lift per unit span on a finite wing must vary in the direction of span, the circulation around the wing must also vary in the direction of span. The total circulation about any section of a wing is the sum of the circulation induced by all of the vortex filaments, which make up the vortex sheet that defines the airfoil section. For an infinite wing, the vortex filaments run in the spanwise direction with no variation in strength along their length, and thus

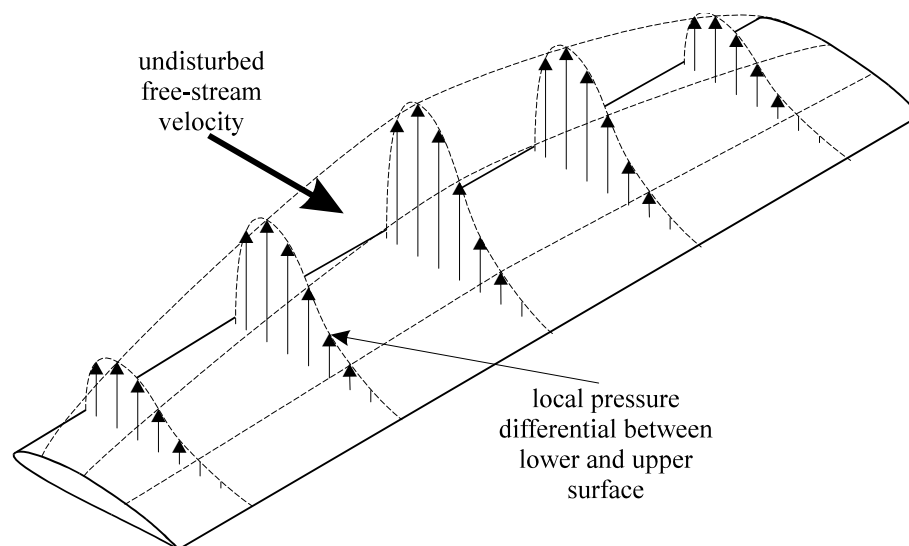


Figure 9.1.1. The lift distribution on a finite wing.

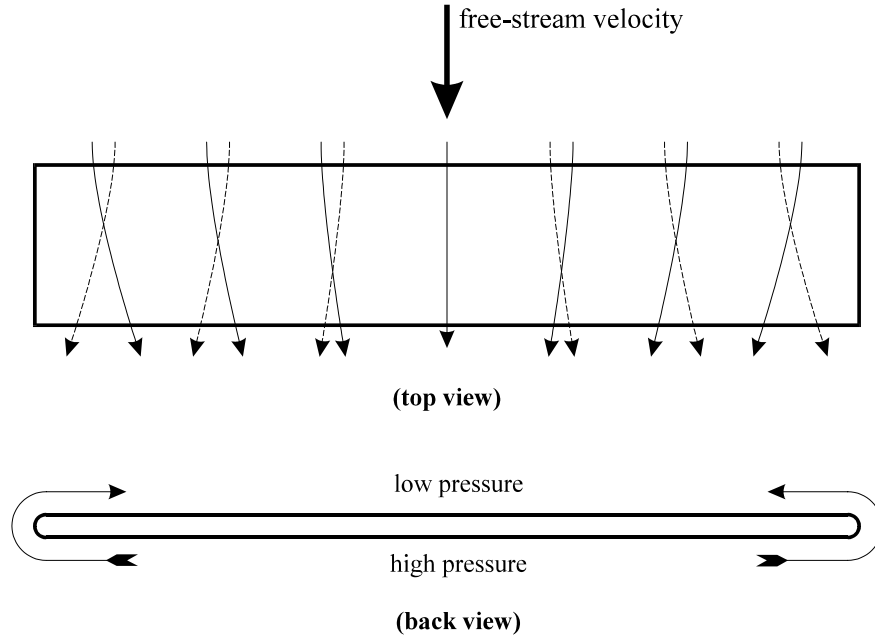


Figure 9.1.2. Airflow around a finite wing.

there is no spanwise variation in lift. For a finite wing, if the circulation and the lift are to vary in the direction of span, then either the strength of the individual vortex filaments must vary along their length or some of the vortex filaments must leave the wing as we move out from the root to the tip. To decide which of these two possibilities is the one that nature dictates, we turn to Helmholtz's vortex theorems.

In the study of two-dimensional potential flow, we found that vortex flow was one solution to the governing differential equations. The two-dimensional vortex, viewed in three dimensions, is simply a straight vortex filament that extends to $\pm\infty$ in the third direction. Other types of vortex filaments are also solutions to the three-dimensional potential flow equations. In general, a three-dimensional vortex filament can be curved in space. However, there are three restrictions that must be placed on a vortex filament if the induced velocity field is to satisfy the three-dimensional potential flow equations.

First, the induced velocity must satisfy what is called the Biot-Savart law (see for example Karamcheti 1966). For an arbitrary vortex filament, such as shown in Fig. 9.1.3, this states

$$d\vec{V} = \frac{\Gamma d\vec{l} \times \vec{r}}{4\pi |\vec{r}|^3} \quad (9.1.1)$$

where $d\vec{V}$ is the differential velocity vector induced at an arbitrary point in space by a differential vector segment of the filament $d\vec{l}$, Γ is the vortex filament strength, and $\vec{r}$ is the radius vector from $d\vec{l}$ to the arbitrary point in space where we wish to compute the induced velocity $d\vec{V}$.

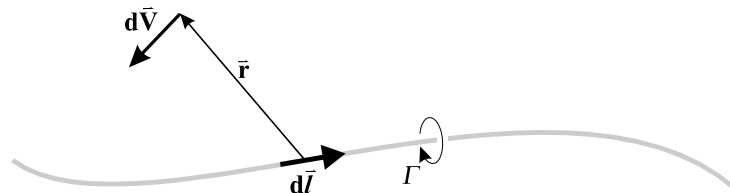


Figure 9.1.3. The velocity induced by a differential segment of a curved vortex filament.

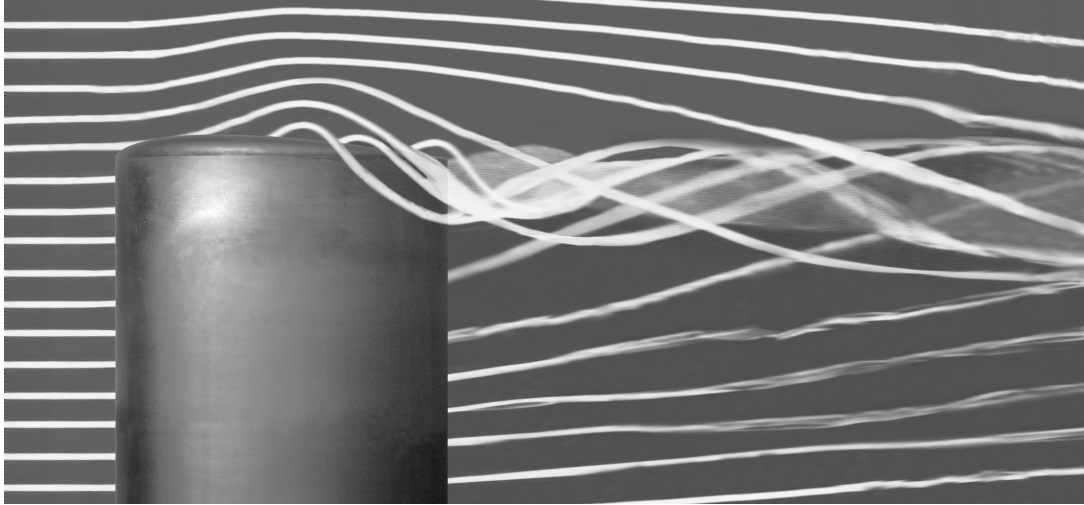


Figure 9.1.4. Vorticity trailing a finite wing.

The other two restrictions that must be placed on the vortex filament are called Helmholtz's vortex theorems. These state

- 1. The strength of a vortex filament in an ideal fluid cannot vary along its length.**
- 2. A vortex filament cannot end in an ideal fluid. It must extend to the fluid boundaries (which can be at $\pm\infty$) or form a closed path.**

These two theorems provide the solution to the mystery of how the circulation about a wing is reduced as we move from the root to the tip. Because the strength of a vortex filament cannot change along its length, some of the vortex filaments must leave the wing as we move out from the root to the tip. Flow visualization experiments confirm this fact. Such experiments (see for example Fig. 9.1.4) show that a sheet of vorticity actually trails downstream from the trailing edge of a finite wing. Because this vortex sheet is free and not bound to the wing surface, the flow field induced by the sheet tends to change the shape of the sheet itself as it moves downstream from the wing. For a finite wing producing positive lift, the trailing vortex sheet induces a circulation that results in upward velocities in the region outboard of the wingtips and downward velocities in the region inboard from the wingtips. Thus, this vortex sheet tends to roll up around an axis trailing behind each wingtip. At some distance behind the wing, the sheet becomes completely rolled up to form two large vortices, one trailing behind each wingtip. For this reason, these vortices are referred to as wingtip vortices, even though they are generated over the full span of the wing. The downward velocity over the wing that is induced by the trailing vortex sheet is called downwash. Potential flow theory predicts that the wingtip vortices generated from this trailing vortex sheet must trail behind the wing for an infinite distance with no reduction in strength. In reality, viscous effects will eventually dissipate the energy in these vortices, but this is a very slow process. These vortices will still have significant energy several miles behind a large aircraft and are of sufficient strength to cause loss of control or structural damage to other aircraft following too closely.

From the above discussion, we see that a wing of finite span cannot be synthesized with a vortex sheet made up of vortex filaments that are always perpendicular to the airfoil sections of the wing. We can, however, still synthesize a finite wing with a vortex sheet, but the sheet must be made up of horseshoe shaped vortex filaments. These filaments run out along the wing in the direction of span and then curves back, eventually leaving the wing from the trailing edge at some point inboard of the wingtip. This can be seen schematically in the plan view of the elliptic wing shown in Fig. 9.1.5. Because the strength of a vortex filament cannot vary along the length of the filament, the decrease in the circulation about the wing as we move out from the root toward the tip must result entirely from the vorticity that leaves the wing in the trailing vortex sheet.

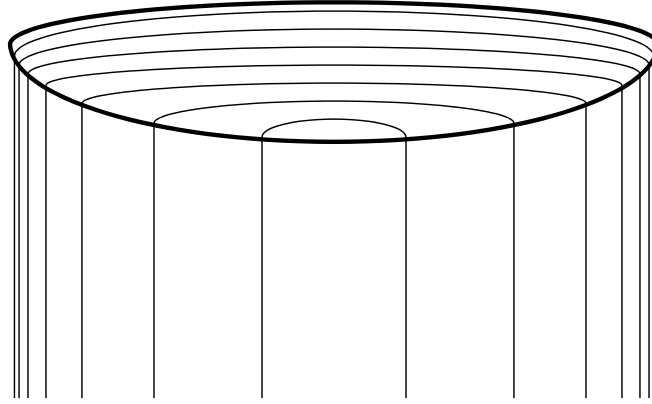


Figure 9.1.5. Schematic of the vorticity distribution on a finite wing with elliptic planform shape.

The classical lifting-line theory, developed by Prandtl (1918), was the first analytical method to satisfactorily predict the performance of finite wings. While a general three-dimensional vortex lifting law was not available at the time of Prandtl's development, the two-dimensional vortex lifting law of Kutta (1902) and Joukowski (1906) was well known. Prandtl hypothesized that each spanwise section of a finite wing has a section lift that is equivalent to that acting on a similar section of an infinite wing having the same section circulation. However, to reconcile this theory with reality, the undisturbed freestream velocity in the Kutta-Joukowski law was intuitively replaced with the vector sum of the freestream velocity and the velocity induced by the trailing vortex sheet. Prandtl's lifting-line theory gives good agreement with experimental data for straight wings of aspect ratio greater than about four. In this section we shall present a development of this very important theory that is still widely used today.

Let x be the axial coordinate measured along the chord in the direction of flow and y be the normal coordinate in the plane of the airfoil sections as has been our usual convention. Then the z -coordinate is in the direction of span from right to left when looking down on the top of the wing, as shown in Fig. 9.1.6. If the origin is in the center of the span, b , then the wing extends from $z = -b/2$ to $z = b/2$. Let the circulation about any section of the wing be $\Gamma(z)$ and let the strength of the trailing vortex sheet per unit span be $\gamma(z)$.

From Helmholtz's first vortex theorem, the decrease in the circulation about the wing with respect to z must equal the strength of the trailing vortex sheet per unit span,

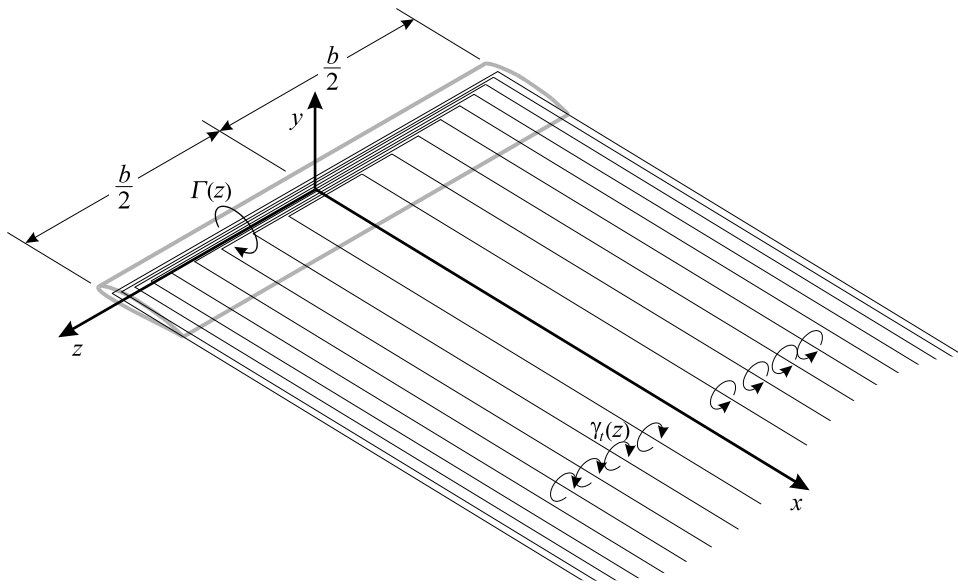


Figure 9.1.6. Prandtl's model for the bound vorticity and the trailing vortex sheet generated by a finite wing.

$$-\frac{d\Gamma}{dz} = \gamma_t(z) \quad (9.1.2)$$

Let the velocity vector that is induced at the point (x,y,z) , by the entire trailing vortex sheet, be $\bar{\mathbf{V}}_t(x,y,z)$. To obtain a mathematical relationship for this induced velocity, we first apply the Biot-Savart law to obtain the differential velocity vector that is induced at the point (x,y,z) by a differential segment of the trailing vortex sheet located at the point $(\xi,0,\zeta)$ and having a length $d\xi$. The directed segment length for this filament, $d\bar{\mathbf{l}}$, is

$$d\bar{\mathbf{l}} = -d\xi\bar{\mathbf{i}}_x + 0\bar{\mathbf{i}}_y + 0\bar{\mathbf{i}}_z \quad (9.1.3)$$

where $\bar{\mathbf{i}}_x$, $\bar{\mathbf{i}}_y$, and $\bar{\mathbf{i}}_z$ are the unit vectors in the x , y , and z directions respectively. The radius vector from the filament at $(\xi,0,\zeta)$ to the point at (x,y,z) is

$$\bar{\mathbf{r}} = (x-\xi)\bar{\mathbf{i}}_x + y\bar{\mathbf{i}}_y + (z-\zeta)\bar{\mathbf{i}}_z \quad (9.1.4)$$

The cross product of $d\bar{\mathbf{l}}$ and $\bar{\mathbf{r}}$ is then

$$d\bar{\mathbf{l}} \times \bar{\mathbf{r}} = \begin{vmatrix} \bar{\mathbf{i}}_x & \bar{\mathbf{i}}_y & \bar{\mathbf{i}}_z \\ -d\xi & 0 & 0 \\ (x-\xi) & y & (z-\zeta) \end{vmatrix} = (z-\zeta)d\xi\bar{\mathbf{i}}_y - yd\xi\bar{\mathbf{i}}_z \quad (9.1.5)$$

The strength of the trailing vortex sheet per unit span is $\gamma_t(z)$. Thus, the total strength of a differential segment of the sheet, located at $z = \zeta$ and having a width $d\xi$, is

$$\Gamma_t(\zeta) = \gamma_t(\zeta) d\xi \quad (9.1.6)$$

Using Eqs. (9.1.4), (9.1.5), and (9.1.6) in Eq. (9.1.1)

$$d\bar{\mathbf{V}}_t = \frac{\Gamma_t d\bar{\mathbf{l}} \times \bar{\mathbf{r}}}{4\pi |\bar{\mathbf{r}}|^3} = \frac{\gamma_t(\zeta)[(z-\zeta)\bar{\mathbf{i}}_y - y\bar{\mathbf{i}}_z]d\xi d\zeta}{4\pi[(x-\xi)^2 + y^2 + (z-\zeta)^2]^{3/2}} \quad (9.1.7)$$

Thus we see that, in the x - z plane, the velocity at the point $(x,0,z)$, induced by a differential segment of the trailing vortex sheet located at the point $(\xi,0,\zeta)$, is parallel to the y axis. If γ_t is positive, the induced velocity is in the positive y direction when z is greater than ζ and in the negative y direction when z is less than ζ .

The total y -velocity component that is induced at the point (x,y,z) by the entire trailing vortex sheet is then

$$V_{ty}(x,y,z) = \frac{1}{4\pi} \int_{\xi=-b/2}^{b/2} \int_{\zeta=0}^{\infty} \frac{d\xi}{[(x-\xi)^2 + y^2 + (z-\zeta)^2]^{3/2}} \gamma_t(\zeta)(z-\zeta)d\zeta \quad (9.1.8)$$

The integration with respect to ξ can be carried out without knowing how the strength of the vortex sheet varies along the span. This integration gives

$$\int_{\xi=0}^{\infty} \frac{d\xi}{[(x-\xi)^2 + y^2 + (z-\zeta)^2]^{3/2}} = \frac{1}{y^2 + (z-\zeta)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (z-\zeta)^2}} \right] \quad (9.1.9)$$

Using Eq. (9.1.9) in Eq. (9.1.8) we have

$$V_{ty}(x,y,z) = \frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{\gamma_t(\zeta)(z-\zeta)}{y^2 + (z-\zeta)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (z-\zeta)^2}} \right] d\zeta \quad (9.1.10)$$

Now, using Eq. (9.1.2) in Eq. (9.1.10), we have

$$V_{ty}(x, y, z) = -\frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{z-\zeta}{y^2+(z-\zeta)^2} \left[1 + \frac{x}{\sqrt{x^2+y^2+(z-\zeta)^2}} \right] d\zeta \quad (9.1.11)$$

We can express the velocity induced by the bound portion of the vortex sheet, $\bar{\mathbf{V}}_b(x, y, z)$, in a similar manner. When computing the velocity induced by the vortex sheet bound to the wing, we approximate the sheet by assuming that all vortices are concentrated along a lifting-line that falls on the z -axis. Again we apply the Biot-Savart law to obtain the differential velocity vector that is induced at the point (x, y, z) by a differential segment of the bound vortex located at the point $(0, 0, \zeta)$ and having a length $d\zeta$. The directed segment length for this filament, $d\bar{\mathbf{l}}$, is

$$d\bar{\mathbf{l}} = 0\bar{\mathbf{i}}_x + 0\bar{\mathbf{i}}_y - d\zeta\bar{\mathbf{i}}_z \quad (9.1.12)$$

The radius vector from the filament at $(0, 0, \zeta)$ to the point at (x, y, z) is

$$\bar{\mathbf{r}} = x\bar{\mathbf{i}}_x + y\bar{\mathbf{i}}_y + (z - \zeta)\bar{\mathbf{i}}_z \quad (9.1.13)$$

The cross product of $d\bar{\mathbf{l}}$ and $\bar{\mathbf{r}}$ is then

$$d\bar{\mathbf{l}} \times \bar{\mathbf{r}} = \begin{vmatrix} \bar{\mathbf{i}}_x & \bar{\mathbf{i}}_y & \bar{\mathbf{i}}_z \\ 0 & 0 & -d\zeta \\ x & y & (z - \zeta) \end{vmatrix} = yd\zeta\bar{\mathbf{i}}_x - xd\zeta\bar{\mathbf{i}}_y \quad (9.1.14)$$

The strength of the bound vortex located at $z = \zeta$ is $\Gamma(\zeta)$. Thus, using Eqs. (9.1.13) and (9.1.14) in Eq. (9.1.1) gives

$$d\bar{\mathbf{V}}_b = \frac{\Gamma d\bar{\mathbf{l}} \times \bar{\mathbf{r}}}{4\pi |\bar{\mathbf{r}}|^3} = \frac{\Gamma(\zeta)[y\bar{\mathbf{i}}_x - x\bar{\mathbf{i}}_y]d\zeta}{4\pi [x^2 + y^2 + (z - \zeta)^2]^{3/2}} \quad (9.1.15)$$

The total y -velocity component induced at the point (x, y, z) by the entire bound vortex sheet is

$$V_{by}(x, y, z) = -\frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{\Gamma(\zeta)xd\zeta}{[x^2 + y^2 + (z - \zeta)^2]^{3/2}} \quad (9.1.16)$$

For a wing generating positive lift, Γ is always positive. Thus, for a lifting wing, we see that the y -component of velocity induced by the bound vortex sheet is negative aft of the wing and positive forward of the wing.

The total y -component of velocity induced by a finite wing at any point, $V_y(x, y, z)$, is the sum of the y -velocity induced by the bound vortex sheet and that induced by the trailing vortex sheet. Combining Eqs. (9.1.11) and (9.1.16), this gives

$$V_y(x, y, z) = -\frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \left\{ \frac{d\Gamma}{d\zeta} \frac{z-\zeta}{y^2+(z-\zeta)^2} \left[1 + \frac{x}{\sqrt{x^2+y^2+(z-\zeta)^2}} \right] + \frac{\Gamma(\zeta)x}{[x^2+y^2+(z-\zeta)^2]^{3/2}} \right\} d\zeta \quad (9.1.17)$$

For a finite wing generating positive lift, the velocity in the y -direction induced over the wing, $V_y(0, 0, z)$, as computed from Eq. (9.1.17), is always negative. Thus, this component of velocity is called the downwash. The total relative wind at any spanwise section of the wing is the vector sum of the freestream velocity and the local downwash, as shown in Fig 9.1.7.

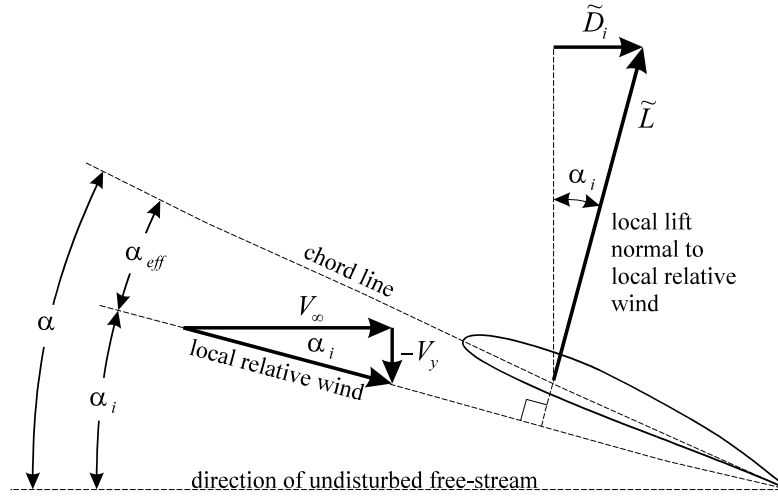


Figure 9.1.7. The induced angle of attack.

Because of the downwash, the local relative wind is inclined at an angle, α_i , to the freestream. This angle is called the induced angle of attack. The downwash is much smaller than the freestream velocity. Thus, the induced angle of attack on the wing is small and we can write

$$\alpha_i(z) = -\frac{V_y(0,0,z)}{V_\infty} = \frac{1}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} \quad (9.1.18)$$

Since the downwash can vary over the span, the induced angle of attack, the effective angle of attack, and the section lift coefficient can also vary in the direction of span. The local section lift coefficient for the airfoil section located at z is

$$\tilde{C}_L(z) = \tilde{C}_{L,\alpha}[\alpha_{eff}(z) - \alpha_{L0}(z)] \quad (9.1.19)$$

where $\tilde{C}_{L,\alpha}$ is the airfoil section lift slope (2π for thin airfoils), α_{eff} is the effective angle of attack, and α_{L0} is the zero lift angle of attack, which is a function only of the shape of the local airfoil section. Since the shape of the local airfoil section could vary along the wing span, as is shown in Fig. 9.1.8, we have allowed the zero lift angle of attack to be a function of z . If a wing has a spanwise variation in the zero lift angle of attack, the wing is said to have aerodynamic twist. Deflecting a trailing-edge flap that extends over only part of the span is one form of aerodynamic twist (see Fig 9.1.9).

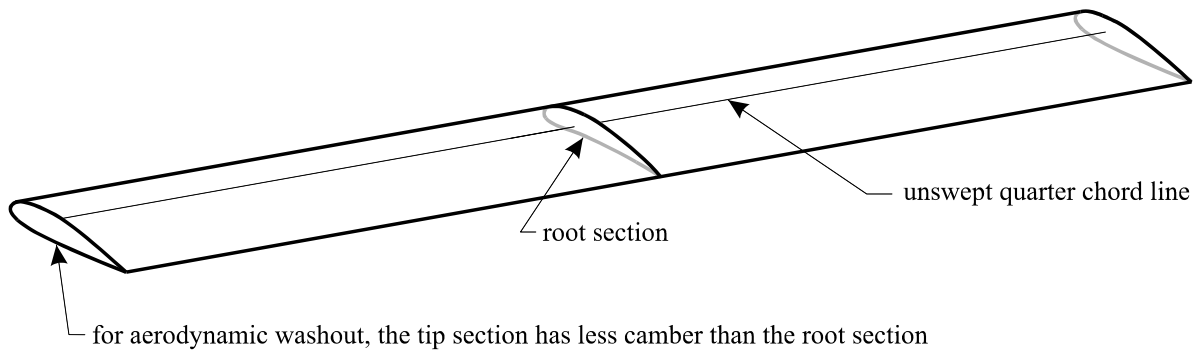


Figure 9.1.8. Aerodynamic twist in an unswept rectangular wing.

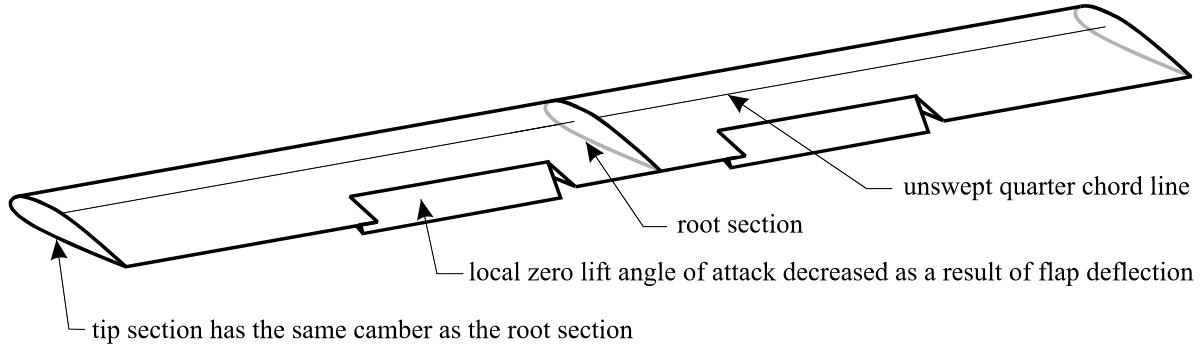


Figure 9.1.9. Aerodynamic twist resulting from deflection of a trailing edge flap spanning only part of the wing.

The effective angle of attack is the geometric angle of attack less the induced angle of attack,

$$\alpha_{eff}(z) = \alpha(z) - \alpha_i(z) \quad (9.1.20)$$

Here we have also allowed the geometric angle of attack, α , to vary in the direction of span, as shown in Fig. 9.1.10. If a wing has a spanwise variation in the geometric angle of attack, the wing is said to have geometric twist. Commonly, the tip is at a lower angle of attack than the root, in which case the geometric twist is called washout. After using Eq. (9.1.20) in Eq. (9.1.19) we have

$$\tilde{C}_L(z) = \tilde{C}_{L,\alpha}[\alpha(z) - \alpha_i(z) - \alpha_{L=0}(z)] \quad (9.1.21)$$

Here we apply Prandtl's original hypothesis. Prandtl assumed that each spanwise section of a finite wing has a section lift that is equivalent to that acting on a similar section of an infinite wing having the same section circulation. On the basis of this hypothesis, Prandtl applied the two-dimensional Kutta-Joukowski law to this three-dimensional potential flow. Since the development and proof of the Kutta-Joukowski law applies only to two-dimensional flows, there was no theoretical basis for Prandtl's hypothesis and no formal proof for his hypothesis was presented. Nevertheless, based on this hypothesis, the local section lift coefficient can also be expressed in terms of the local circulation. Using the definition of section lift coefficient and the Kutta-Joukowski law we have

$$\tilde{C}_L(z) = \frac{\tilde{L}}{\frac{1}{2}\rho V_\infty^2 c} = \frac{\rho V_\infty \Gamma}{\frac{1}{2}\rho V_\infty^2 c} = \frac{2\Gamma(z)}{V_\infty c(z)} \quad (9.1.22)$$

Since most finite wings have a variable chord, we also allow the chord length to vary in the direction of span.

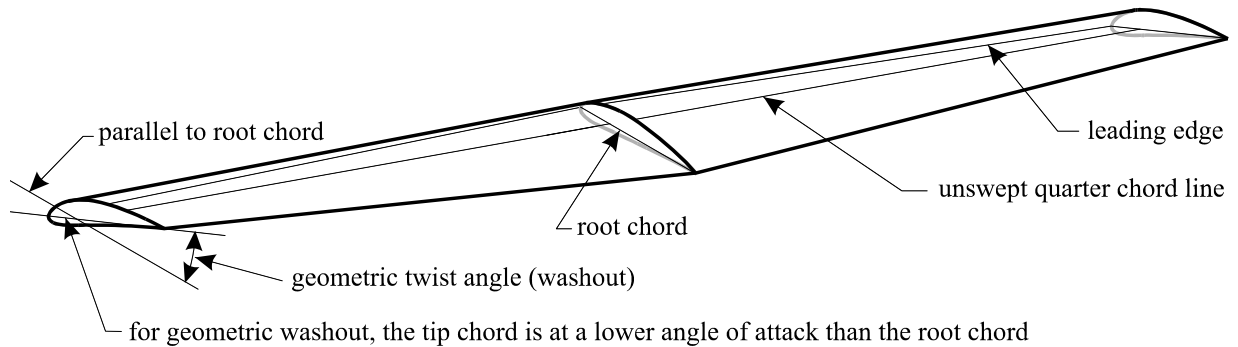


Figure 9.1.10. Geometric twist in an unswept rectangular wing.

Combining Eqs. (9.1.21) and (9.1.22) to eliminate the section lift coefficient we have

$$\frac{2\Gamma(z)}{V_\infty c(z)} = \tilde{C}_{L,\alpha}[\alpha(z) - \alpha_i(z) - \alpha_{L=0}(z)] \quad (9.1.23)$$

Using Eq. (9.1.18) in Eq. (9.1.23) and rearranging gives

$$\frac{2\Gamma(z)}{V_\infty c(z)} + \frac{\tilde{C}_{L,\alpha}}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} = \tilde{C}_{L,\alpha}[\alpha(z) - \alpha_{L=0}(z)] \quad (9.1.24)$$

Equation (9.1.24) is the fundamental equation of Prandtl's lifting-line theory. It is a non-linear integro-differential equation that involves only one unknown, the local circulation as a function of the spanwise position, $\Gamma(z)$. All other parameters are known, for a given wing design at a given geometric angle of attack and a given freestream velocity.

Once the circulation distribution has been determined from Eq. (9.1.24), based on Prandtl's hypothesis, the section lift distribution in the spanwise direction can be obtained from the two-dimensional Kutta-Joukowski law,

$$\tilde{L}(z) = \rho V_\infty \Gamma(z) \quad (9.1.25)$$

Again based on Prandtl's hypothesis, the local lift vector should be perpendicular to the local relative wind and inclined at an angle, α_i , to the free stream, as shown in Fig 9.1.7. Since the induced angle of attack is small, the total lift can be approximated as

$$L = \int_{z=-b/2}^{b/2} \cos(\alpha_i) \tilde{L}(z) dz \cong \int_{z=-b/2}^{b/2} \tilde{L}(z) dz = \rho V_\infty \int_{z=-b/2}^{b/2} \Gamma(z) dz \quad (9.1.26)$$

and the total lift coefficient for a wing is found to be

$$C_L = \frac{L}{\frac{1}{2} \rho V_\infty^2 S_p} = \frac{2}{V_\infty S_p} \int_{z=-b/2}^{b/2} \Gamma(z) dz \quad (9.1.27)$$

where S_p , is planform area of the wing.

Since the local lift vector has a component in the direction of the freestream that makes a contribution to the drag. This can be seen in Fig. 9.1.7. The component of drag, which results from the downwash, is called induced drag. The local section induced drag at any spanwise section of the wing is

$$\tilde{D}_i = \tilde{L} \sin(\alpha_i) \quad (9.1.28)$$

or, since the induced angle of attack is small, and in view of Eqs. (9.1.18) and (9.1.25)

$$\tilde{D}_i(z) \cong \tilde{L}(z) \alpha_i(z) = \rho V_\infty \Gamma(z) \alpha_i(z) = \frac{\rho \Gamma(z)}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} \quad (9.1.29)$$

Thus, the total induced drag is

$$D_i = \int_{z=-b/2}^{b/2} \tilde{D}_i(z) dz = \rho V_\infty \int_{z=-b/2}^{b/2} \Gamma(z) \alpha_i(z) dz = \int_{z=-b/2}^{b/2} \frac{\rho \Gamma(z)}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} dz \quad (9.1.30)$$

and the total induced drag coefficient is

$$C_{D_i} = \frac{D_i}{\frac{1}{2}\rho V_\infty^2 S_p} = \frac{2}{V_\infty S_p} \int_{z=-b/2}^{b/2} \Gamma(z) \alpha_i(z) dz = \int_{z=-b/2}^{b/2} \frac{\Gamma(z)}{2\pi V_\infty^2 S_p} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} dz \quad (9.1.31)$$

Thus Prandtl's lifting-line theory allows us to predict how the downwash induced by the vortex sheet trailing from a finite wing will affect the lift and drag on the wing itself. This downwash will also affect the aerodynamic forces and moments on other surfaces of the aircraft such as the tail and fuselage.

In **summary**, Prandtl's lifting-line theory allows us to compute the local circulation, Γ , as a function of the spanwise position along the wing, z . For a given wing design, at a given angle of attack, the chord length, $c(z)$, the geometric angle of attack $\alpha(z)$, and the zero-lift angle of attack, $\alpha_{L=0}(z)$ are all known functions of the spanwise position. Then, for a given freestream velocity, the **lifting-line equation** provides a single non-linear integro-differential equation, in which the only unknown is the local circulation, $\Gamma(z)$.

$$\frac{2\Gamma(z)}{V_\infty c(z)} + \frac{\tilde{C}_{L,\alpha}}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} = \tilde{C}_{L,\alpha} [\alpha(z) - \alpha_{L=0}(z)] \quad (9.1.24)$$

Once the circulation distribution, $\Gamma(z)$, has been determined from this equation, we can compute the **downwash** at any point (x, y, z) that is induced by the lifting wing,

$$V_y(x, y, z) = -\frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \left\{ \frac{d\Gamma}{d\zeta} \frac{z-\zeta}{y^2 + (z-\zeta)^2} \left[1 + \frac{x}{\sqrt{x^2 + y^2 + (z-\zeta)^2}} \right] + \frac{\Gamma(\zeta)x}{[x^2 + y^2 + (z-\zeta)^2]^{3/2}} \right\} d\zeta \quad (9.1.17)$$

the **induced angle of attack** at any point along the span of the wing,

$$\alpha_i(z) = -\frac{V_y(0, 0, z)}{V_\infty} = \frac{1}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} \quad (9.1.18)$$

the **lift coefficient**,

$$C_L = \frac{2}{V_\infty S_p} \int_{z=-b/2}^{b/2} \Gamma(z) dz \quad (9.1.27)$$

and the **induced drag coefficient**,

$$C_{D_i} = \frac{2}{V_\infty S_p} \int_{z=-b/2}^{b/2} \Gamma(z) \alpha_i(z) dz = \int_{z=-b/2}^{b/2} \frac{\Gamma(z)}{2\pi V_\infty^2 S_p} \int_{\zeta=-b/2}^{b/2} \frac{d\Gamma}{d\zeta} \frac{d\zeta}{z-\zeta} dz \quad (9.1.31)$$

From Prandtl's lifting-line theory we have seen that a finite lifting wing results in the generation of downwash. This downwash has a significant effect on the performance of the wing and other surfaces of an airplane. Mathematically we have described the downwash and its effects in terms of vorticity and circulation. However, the student may find it easier to understand the physics of downwash when it is described in terms of pressure and momentum.

When an airplane flies overhead, the downwash produced by the lifting wing could be measured on the ground, if instruments of sufficient sensitivity were available. This downwash can be viewed as a result the momentum imparted to the fluid by the force of the airplane's weight. The weight of the airplane is directly carried by the

pressure difference between the upper and lower surfaces of the wing. However, the fluid also feels this same pressure difference. The net effect of the pressure difference on the fluid is that a force is applied to the fluid equal to the airplane's weight. From Newton's second law, we know that such a force must result in a downward momentum imparted to the fluid. As the fluid directly beneath the airplane moves down, viscous forces from the adjacent fluid may slow its downward motion. However, these same viscous force acts on the adjacent fluid to impart downward momentum to that fluid as well. Newton's second law requires that the total downward momentum of the fluid cannot change, unless acted upon by an external force. Ultimately the earth will stop this downward fluid momentum, which began as the downwash from the airplane wing. The final manifestation of the downwash produced by a lifting wing is a very small pressure rise over a very large area of the earth. The net effect of this pressure rise integrated over the total affected area is a force applied to the surface of the earth equal to the airplane weight. Whether the airplane is resting on the ground or flying at 40,000 feet, the surface of the earth must ultimately support the total weight of the aircraft. Because the airplane is moving rapidly over the ground, the downwash is spread over a very large area of the earth and usually goes undetected.

The vortex model that we have used to describe the effects of downwash on a finite wing is based on inviscid flow. This model predicts that the vorticity shed from the wing and the associated downwash will extend an infinite distance behind the wing. In reality viscous effects in the air will eventually dissipate the wingtip vortices. However, the net downward momentum associated with these vortices does not change as a result of this viscous dissipation. No matter how large the viscosity, this downward momentum will eventually reach the ground, where it produces an aerodynamic force equal to the airplane's weight.